ENGINE B

The Forecast Engine: how it works and how to build the best one

From the demo on the concept page to a real predictor for China → Algeria freight that you can measure, trust and sell.

1 The demo, step by step
2 What moves freight prices
3 Building it in 4 levels
4 Measuring the best one
5 A better decision rule
6 What to start this week

1 What the demo does, step by step

Worked example: **Shanghai → Algiers, 40' container, shipping in 3 weeks (around 21 Nov).**

STEP 1
BASE PRICE

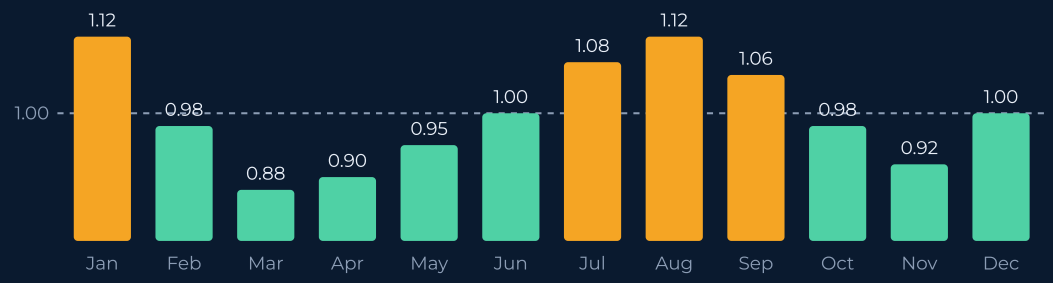
Every route starts from a base price

3,450 USD (40' base) **× 1.00** (Shanghai) **× 1.00** (Algiers) **=** 3,450 USD

Each port has a small factor. Qingdao is ×1.03 (slightly further). Annaba is ×1.08 (fewer ships call there).

STEP 2
MONTH
FACTOR

Every month has a number for how expensive freight usually is



- **January is high**: everyone ships before Chinese New Year.
- **March is low**: factories are restarting after the holiday.
- **July to September** is peak season.

21 Nov sits between Nov (0.92) and Dec (1.00) → blended factor 0.973
3,450 × 0.973 = 3,358 USD

STEP 3
WIGGLE

A small random movement each week

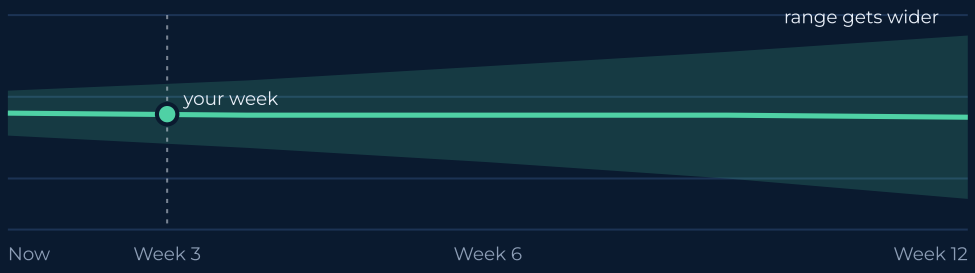
Added so the line looks like a real market. **This part is fake.** It's there to make the picture look realistic, and it predicts nothing.

STEP 4
THE RANGE

The shaded area: where the price will probably land

Range = expected price ± (9% + 1.2% per week ahead)

3 weeks out: ±12.6% → about 2,900 to 3,740 USD



Nobody knows the price 12 weeks from now as well as next week's, so the range widens the further out you look.

STEP 5
THE SIGNAL

Compare your week with 2 weeks later

PRICE 2 WEEKS AFTER YOUR DATE	SIGNAL
More than 3% higher	**Book now**
More than 3% lower	**Wait**
Anything in between	**Stable**

THE HONEST SUMMARY

The demo is a toy. I estimated the month factors, the wiggle is random, and the 9% range is a guess. It shows what the engine does, not how a real one does it. The rest of this guide is how to build the real one.

2 What actually moves freight prices

To predict a price, you need to know what pushes it up and down.

DRIVER	EFFECT	HOW FAR AHEAD YOU CAN SEE IT
Season Chinese New Year, peak season, Golden Week (early October)	Repeats every year	Months ahead
Carrier announcements GRI (general rate increase), PSS (peak season surcharge)	Carriers announce price rises in advance	**2–4 weeks ahead.** Your strongest short-term signal.
Blank sailings	Carriers cancel voyages to reduce space and push prices up	2–6 weeks ahead
Fuel price	Feeds the BAF (fuel surcharge). Rises and falls with oil.	Weeks
Disruptions Red Sea / Suez, strikes, port congestion	Sudden jumps. When ships avoided the Red Sea in 2024, voyages got 10–14 days longer and prices multiplied.	Not predictable. Only detectable fast.
Algeria-specific	Transshipment through Med hubs, port congestion, import rules. Algeria pays a premium on top of the main Mediterranean price.	Measured from real quotes

PREDICTABLE

Season, announced GRIs and PSS, blank sailings. **The engine uses these to forecast.**

NOT PREDICTABLE

Wars, canal closures, strikes. **The engine can't forecast these.** It can detect them quickly and widen the range.

Key vocabulary

Index	A published weekly average freight price for a main route (SCFI, Drewry WCI, Freightos FBX).
Spot rate	The price for a shipment booked now, as opposed to a long-term contract price.
Lane premium	How much more (or less) your exact route costs than the main index route. Algiers vs Genoa, for example.
Backtest	Testing a forecast method on past weeks to see how accurate it would have been.

3 How to build the real one, in 4 levels

LEVEL 1 · START FROM TODAY'S REAL PRICE

The best simple prediction of next week's price is this week's price.

Professionals call this the **naive forecast**. It's surprisingly hard to beat. So the real engine starts from the latest real price and only adjusts it:

Forecast (target week) =
Today's index price
× Season change (target week vs this week)
× Algeria premium

Weekly index
SCFI (Shanghai–Mediterranean, every Friday), Drewry WCI (Shanghai–Genoa), Freightos FBX (China–Med). Headline numbers are public. Check how much past data is free, because some sources charge for full history.

Algeria premium
No index covers Algeria. Compare real Algiers quotes with the Med index for the same week.
Premium = Algiers quote ÷ Med index
1.18 = Algiers costs 18% more.

Real season factors
Calculate from 5+ years of history instead of guessing. Leave out 2020–2022 (COVID chaos). Use medians so one crazy year doesn't distort everything.

LEVEL 2 · ADD THE EARLY-WARNING SIGNALS

Collect every week: **announced GRIs and PSS** (on carriers' websites and customer notices), **blank sailings**, **fuel price**, and a **disruption flag** (Red Sea status, strikes).

Then train a model to predict **the % change** in the next 2, 4 and 8 weeks, not the price itself. Predicting the change works better because prices jump between very different levels over the years.

- **Start with a simple regression**, because you can explain it to clients.
- **Later, try a tree model like LightGBM**, which is good at "if this and that" patterns.

LEVEL 3 · YOUR OWN DATA, WHICH IS YOUR MOAT

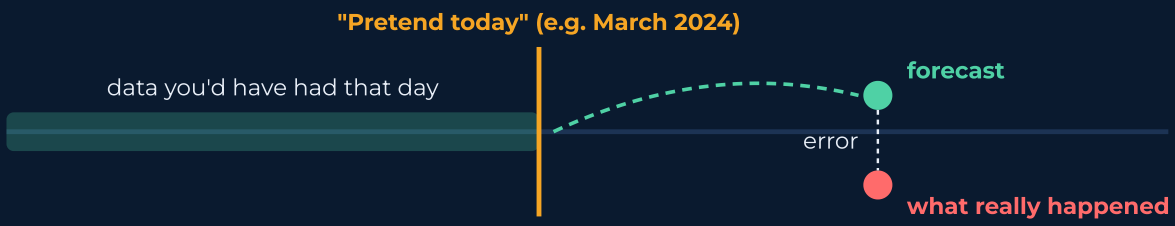
Every user of the free calculator can enter their real quote. After a few hundred quotes you'll know:

- the real Algeria premium **per port** (Algiers vs Oran vs Djen Djen), updated weekly
- which freight forwarders charge above the market

Nobody else has this data for North African routes. A competitor can copy your formulas but can't copy your data.

LEVEL 4 · AN HONEST RANGE, BASED ON YOUR REAL PAST ERRORS

Instead of guessing "±9%", you measure it by **backtesting**:



Repeat for every past week → you get hundreds of real errors.

1. Pretend it's a past week. Using only the data you'd have had that day, make your forecast.
2. Compare it with what the price actually did.
3. Repeat for every week of history.
4. Look at all those errors. If 80% of them fall within ±11% at 3 weeks out, then **your range at 3 weeks is ±11%**.

The range is then based on what actually happened, not on a guess.

The 4 levels at a glance

LEVEL	WHAT IT ADDS	NEEDS
1	Real current price + real seasons + Algeria premium	Weekly index, a few real quotes
2	Early-warning signals, % change model	GRI/PSS, blank sailings, fuel, disruptions
3	Premium per port, forwarder benchmark	Quotes from calculator users
4	Honest range from real past errors	History + backtest

4 How to know you have the best one

Measure these three things on your backtest.

METRIC	QUESTION IT ANSWERS	TARGET
Average % error	How far off is the expected price, on average?	Lower than the naive forecast
Direction hit rate	When it says "up", does the price go up?	Well above 50%. **The one that matters for Book now / Wait.**
Range check	Does the 80% range really contain 80% of real prices?	75–85%

GOLDEN RULE

If your model can't beat **"next week = this week"**, don't ship it. Use the naive forecast plus seasonality instead.

MARKETING BONUS

Publish your accuracy: **"Last 12 weeks: direction right 9 times out of 12."** No competitor shows this. Skeptical buyers trust it more than any slogan.

5 A better decision rule

"+3% in 2 weeks" is simplistic. A real rule asks three questions:

1 · How likely?

Say "Wait" only if the chance of a drop is above 60%.

2 · How much?

Compare the possible saving with the risk of the price rising instead.

3 · Can you wait?

If the goods must arrive by a fixed date, it's "Book now" whatever the market does.

Forwarder quotes are usually valid for about 2 weeks. "Wait" only helps if you can come back for a new quote in time.

6 What you can start this week, for free

A 10-minute routine every Friday, in a Google Sheet:

DATE	SCFI MED	WCI SHANGHAI–GENOA	YOUR ALGIERS QUOTE	GRI ANNOUNCED?	NOTE
Fri 6 Nov	…	…	if any	yes / no	Red Sea, strike…

After 8–12 weeks: your own Algeria premium. **After 6 months:** enough for a backtest, then the Level 1 model in Python (Colab).

Numbers in the demo examples are illustrative, not quotes. Verify index availability and data licensing before using any source commercially.